

1	2	3	4	5	6	Total
/20	/20	/20	/20	/20	/20	/100

Name: _____

June 30, 2016 Thursday

Department/School: _____

SID: _____

University Physics

Final Examination

Kuang Yaming Honors School, Nanjing University

$$\Gamma(\nu) = \int_0^{+\infty} e^{-x} x^{\nu-1} dx, \quad \Gamma(\nu+1) = \nu\Gamma(\nu), \quad \Gamma(1) = 1, \quad \Gamma(1/2) = \sqrt{\pi}.$$

Select five out of following six problems.

1.(20pts.) A spaceship, moving away from the earth at a speed of $0.90c$, reports back by transmitting on a frequency (measured in the spaceship) of 100 MHz.

(a) Will the signal be red or blue shifted when it is received on the earth?

(b) What frequency the earth receiver must be tuned to for receiving this signal?

2. (20pts.) A particle moves on the xy -plane with a speed v at an angle θ with respect to the x -axis ($-\pi < \theta < \pi$, $\theta \neq 0$).

(a) At what speed v' will the particle be observed in the frame S' which is moving along the x -axis with the speed $u = v$.

(b) What is the corresponding angle θ' of the velocity of the particle in the frame S' ?

3. (20pts.) A gas of one mole satisfies the equation of state $pv = RT(1 + Bp)$ where p, v and T are the pressure, molar volume and temperature respectively, R the so-called gas constant, and B a function of T .

(a) Find the coefficient of thermal expansion $\alpha = 1/V (\partial V / \partial T)_p$ of the gas.

(b) Find the isothermal compressibility $\kappa = -1/V (\partial V / \partial p)_T$ of the gas.

(c) Show that both α and κ approach the values of the ideal gas as the pressure p tends to zero.

4. (20pts.) Consider an ideal gas. Suppose its total amount is one mole.

- (a) Show that $ds = c_v dT/T + R dv/v$ where s, T and v are the molar entropy, temperature and molar volume of the ideal gas respectively, R the gas constant, c_v the molar heat capacity at constant volume.
- (b) The ideal gas undergoes a (polytropic) process such that $pv^n = C$ where n and C are both constant ($n \geq 0, n \neq 1$). Find Δs , the change in molar entropy, when the system changes from the initial volume v_1 to the final volume v_2 in this process. Suppose c_v is a constant.
- (c) Find the heat capacity c_n of the ideal gas for the process that is described in (b).

5. (20 pts.) Given a gas with the total number of particles fixed.

- (a) Prove that $(\partial T / \partial p)_H = -1/C_p (\partial H / \partial p)_T$ where T, p and H are the temperature, pressure and enthalpy of the system respectively, and C_p the heat capacity at constant pressure (Hint: The partial derivative $(\partial H / \partial p)_T$ implies that we can take H as a function of T and p , i.e., $H = H(T, p)$).
- (b) Show from $H = G + TS$ that $(\partial T / \partial p)_H = 1/C_p [T(\partial V / \partial T)_p - V]$ where G, S and V are the Gibbs free energy, entropy and volume of the system respectively.
- (c) Find $(\partial T / \partial p)_H$ for the ideal gas.

6. (20pts.) The Maxwell speed distribution has the form $dn = C \exp(-mv^2/2k_B T) v^2 dv$ where m denotes the mass of the molecule of the gas, T the temperature, and k_B the Boltzmann constant. The coefficient C is a constant. We denote $\varepsilon = (1/2)mv^2$ as the energy of the molecule of the gas.

(a) Calculate the statistical average of the speed v , i.e., $\langle v \rangle$.

(b) Calculate $\langle v^2 \rangle$ and $\langle \varepsilon \rangle$.

(c) Calculate $\langle \varepsilon^2 \rangle$.

(d) The uncertainty (or fluctuation) ΔO of a quantity O is defined as $\Delta O = \sqrt{\langle O^2 \rangle - \langle O \rangle^2}$. Find the uncertainties Δv and $\Delta \varepsilon$.